Project Presentation

On

Machine Learning based Student Performance Prediction

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INTRODUCTION

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“ **Student academic performance**” is an important indicator of educational success. Traditional evaluation methods mainly depend on final examinations, which do not help in identifying weak students at an early stage. Because of this delay, educators are unable to provide timely support and intervention.

This research focuses on predicting student academic performance using machine learning techniques. By analyzing factors such as attendance, internal marks, study hours, and previous academic records, the system can identify students who may struggle academically. The objective of this research is to develop an accurate and reliable prediction model that helps institutions improve academic outcomes and reduce dropout rates through early intervention.

APPROACH

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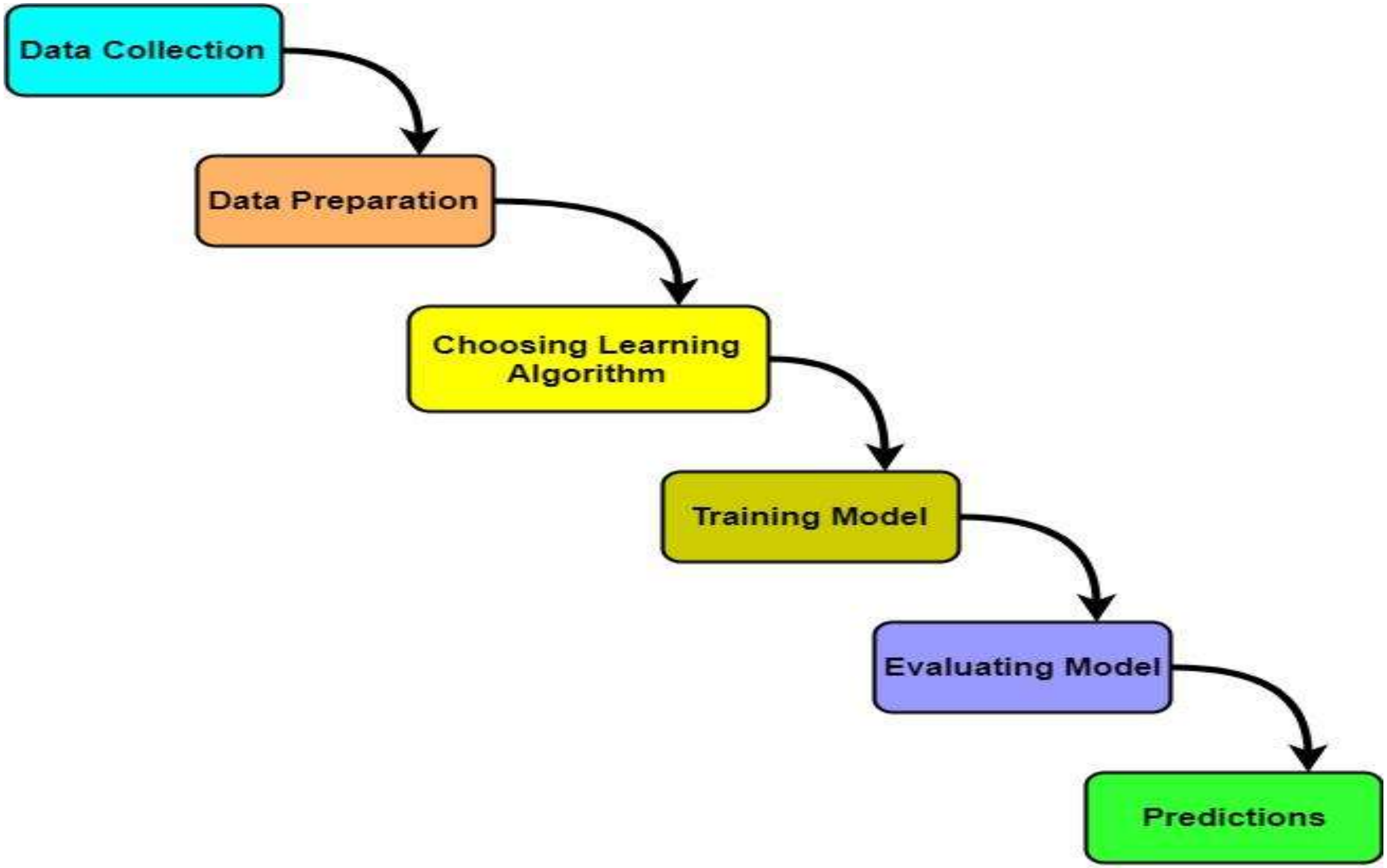
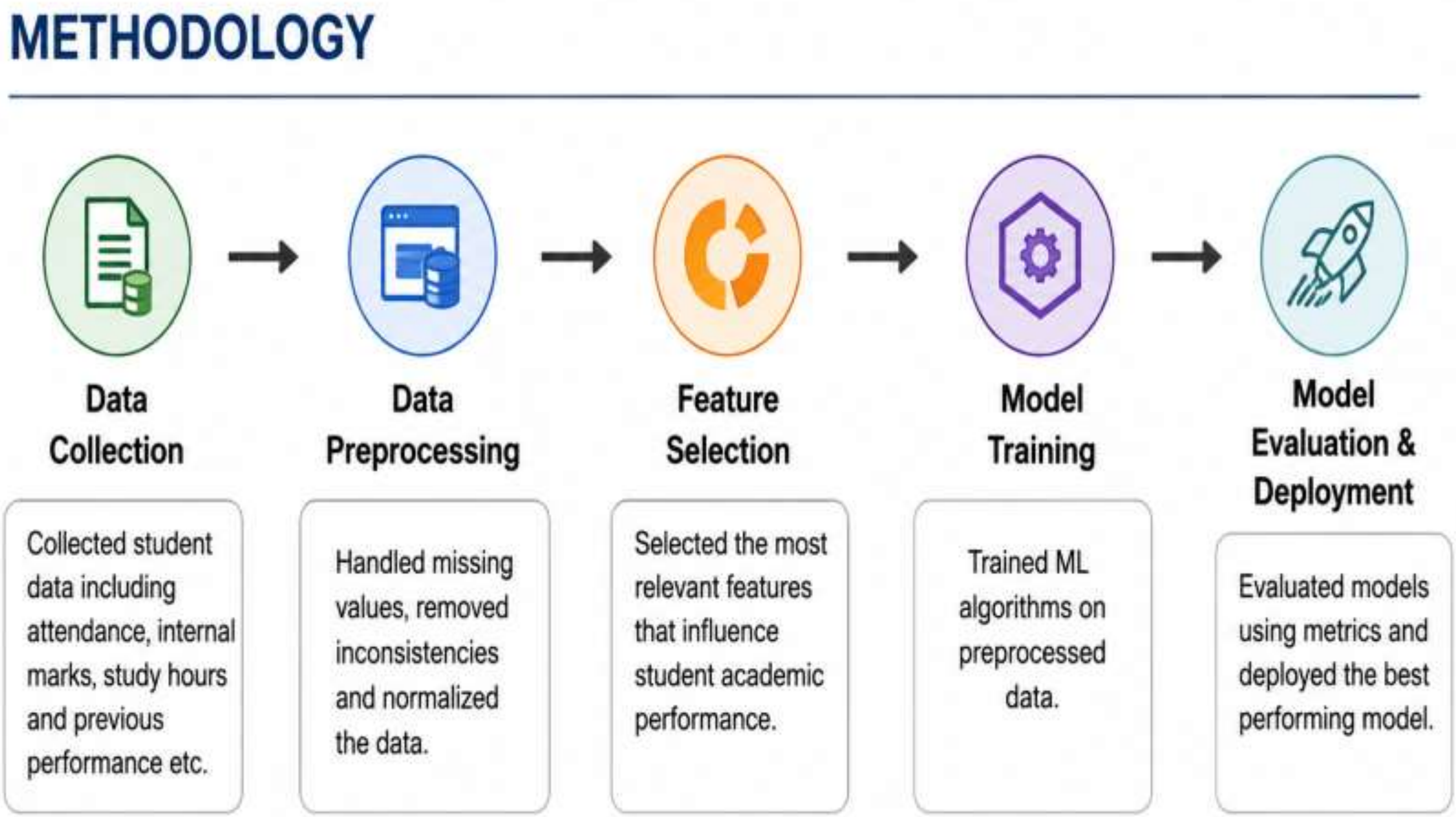
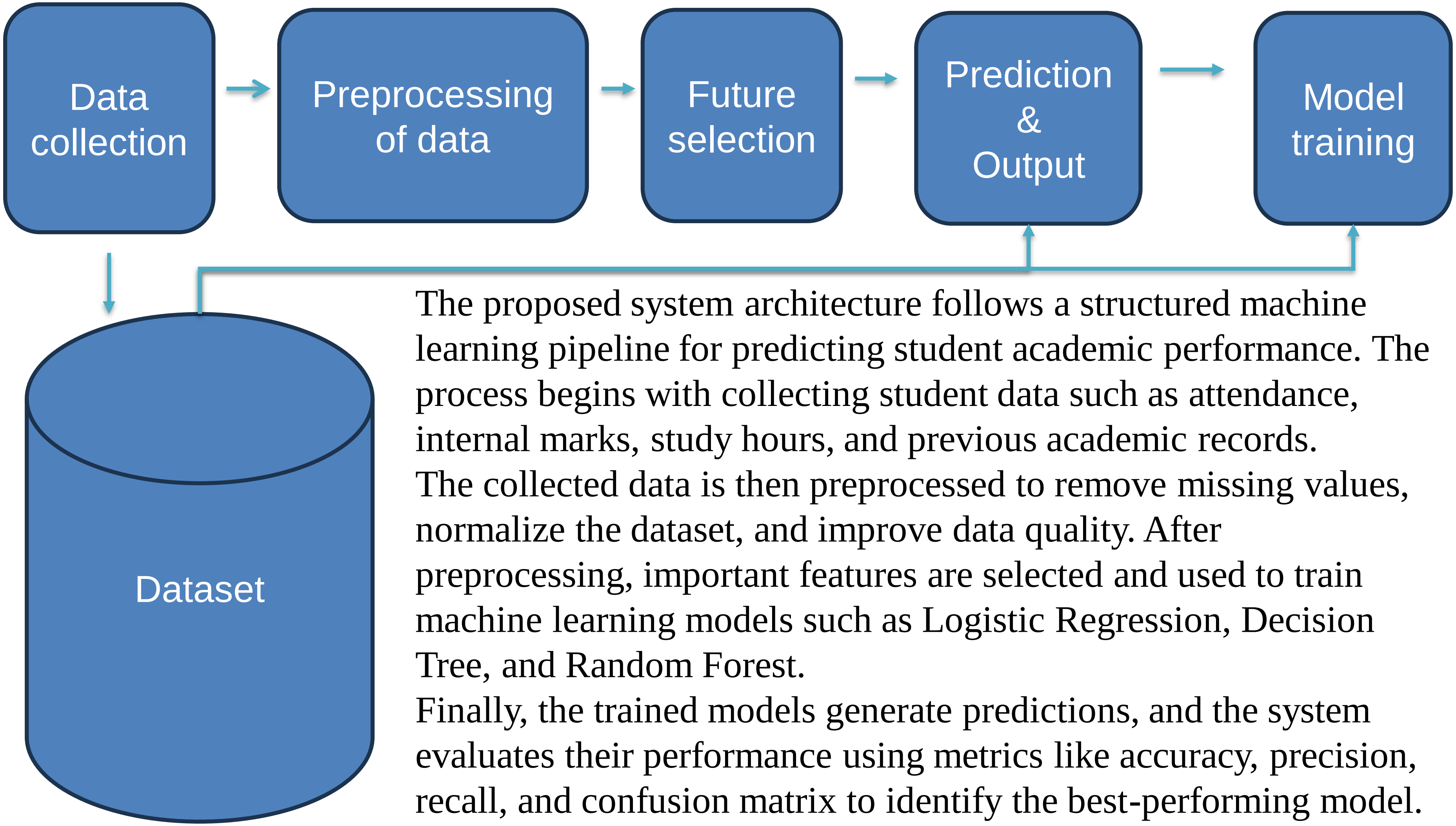


Machine Learning Workflow

METHODOLOGY



SYSTEM ARCITECTURE



The proposed system architecture follows a structured machine learning pipeline for predicting student academic performance. The process begins with collecting student data such as attendance, internal marks, study hours, and previous academic records.
The collected data is then preprocessed to remove missing values, normalize the dataset, and improve data quality. After preprocessing, important features are selected and used to train machine learning models such as Logistic Regression, Decision Tree, and Random Forest.
Finally, the trained models generate predictions, and the system evaluates their performance using metrics like accuracy, precision, recall, and confusion matrix to identify the best-performing model.

EXPERIMENTAL SETUP

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Platform/Language: Python

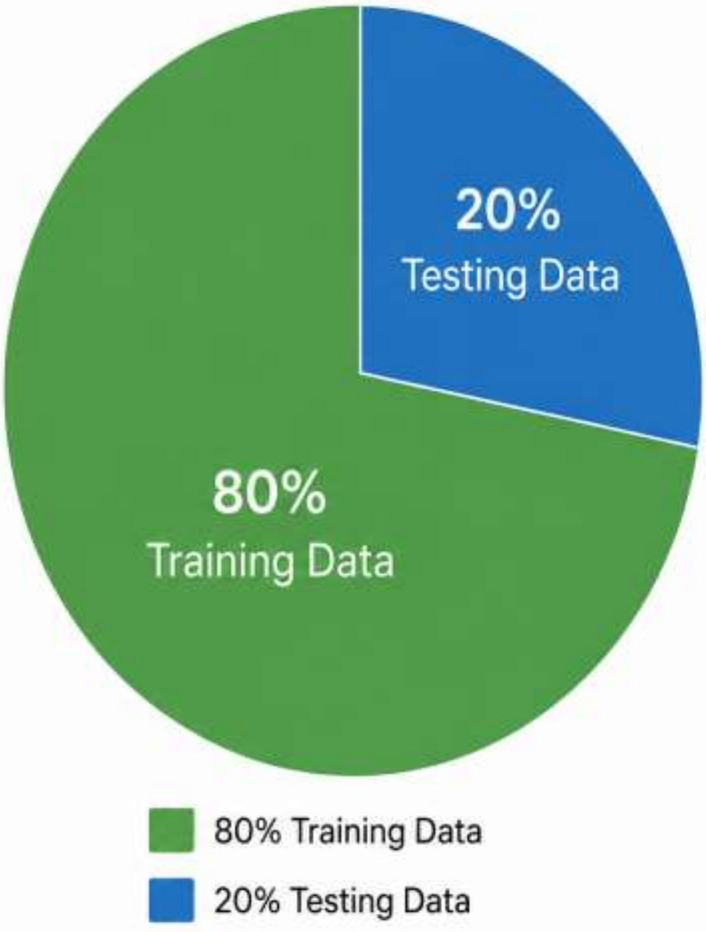
Libraries: Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn

Dataset: Student Academic Performance Dataset

Train-Test Split:

- Training Data – 80%
- Testing Data – 20%

Evaluation Metrics: Accuracy, Precision, Recall, Confusion Matrix



Result and Discussion

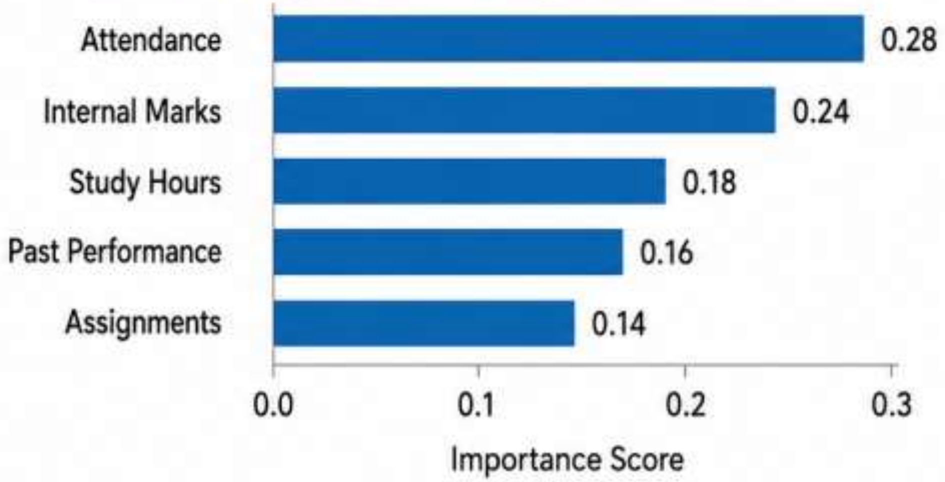
Model Performance Comparison

Algorithm	Accuracy	Precision	Recall	F1-Score
Logistic Regression	78.12%	77.45%	76.80%	77.12%
Decision Tree	83.21%	82.30%	82.10%	82.20%
Random Forest	88.35%	88.10%	88.00%	88.05%

Confusion Matrix (Random Forest)

Actual \ Predicted	Good	Average	Poor
Good	186	12	2
Average	15	142	13
Poor	3	11	115

Feature Importance (Top 5)



Applications and Advantages

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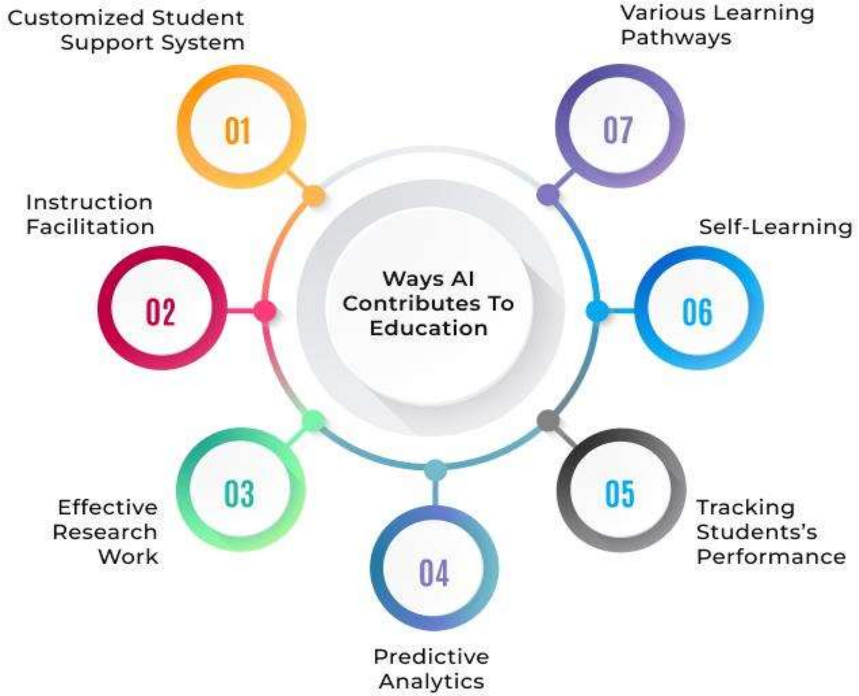
Applications:

1. Early identification of weak students
2. Personalized learning support
3. Better academic planning
4. Career guidance and counseling
5. Real-time student monitoring

Advantages:

1. High accuracy and reliability
2. Early intervention support
3. Scalable for large datasets
4. Objective and unbiased evaluation
5. Easy integration with existing systems



Conclusion

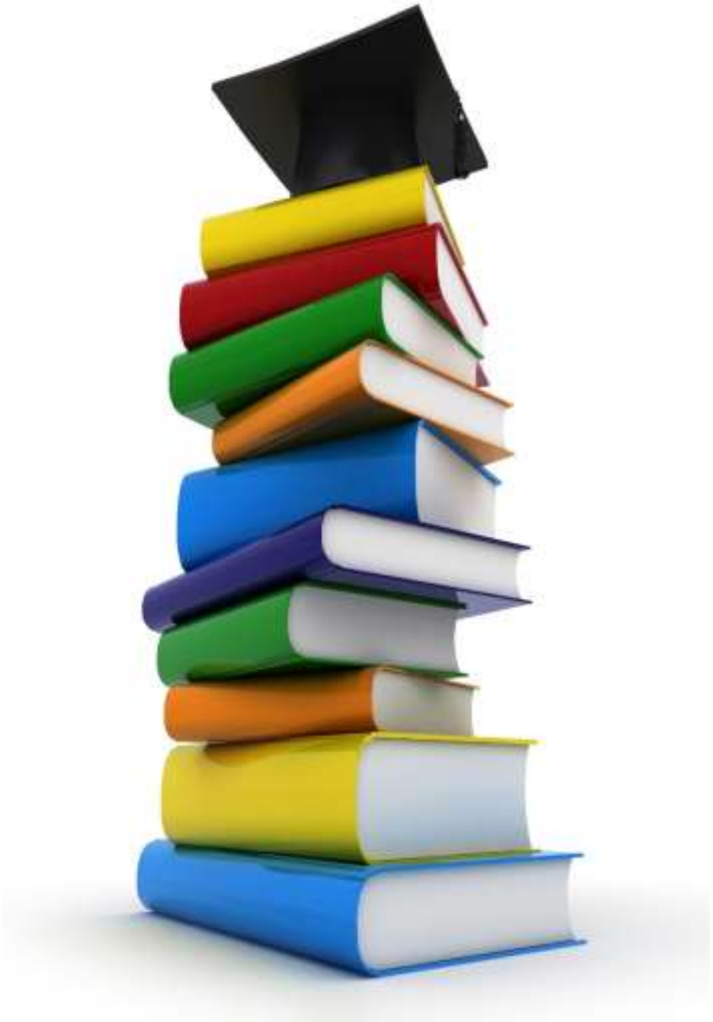
•Machine learning techniques can effectively predict student academic performance.

•Random Forest achieved the highest accuracy (~88%).

•Early prediction helps educators take proactive actions.

•The system improves student success and reduces dropout rates.

•Future enhancements can make the system more intelligent and practical.



Future Scope

Future Improvements

Include psychological and socio-economic factors

Use Deep Learning models

Develop real-time prediction systems

Create mobile and web applications

Integrate with LMS and e-learning platforms

References

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Thank You